EXPERIMENT-6

6. Scenario: You are a cashier at a grocery store and need to calculate the total cost of a customer's purchase, including applicable discounts and taxes. You have the item prices and quantities in separate lists, and the discount and tax rates are given as percentages. Your task is to calculate the total cost for the customer.

Question: Use arithmetic operations to calculate the total cost of a customer's purchase, including discounts and taxes, given the item prices, quantities, discount rate, and tax rate?

Aim:

To calculate the total purchase cost, including discount and tax, using arithmetic operations in Python.

Algorithm:

1. Define item prices, quantities, discount rate, and tax rate.
2. Calculate the subtotal using prices and quantities.
3. Calculate and subtract the discount.
4. Calculate tax on the discounted amount.
5. Display the final cost.

Code:

```
prices = [100, 200, 150]

quantities = [2, 1, 3]

discount_rate = 10

tax_rate = 5

subtotal = sum(prices[i] * quantities[i] for i in range(3))

discount = subtotal * discount_rate / 100

amount = subtotal - discount
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tax = amount * tax_rate / 100

total = amount + tax

print("Total Cost:", total)

Input:

Prices = [100, 200, 150]

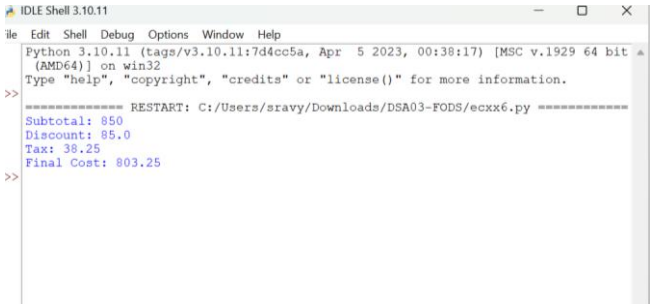
Quantities = [2, 1, 3]

Discount = 10%

Tax = 5%

Output:

Total Cost: 803.25



Result:

The total purchase cost was calculated successfully as ₹803.25.